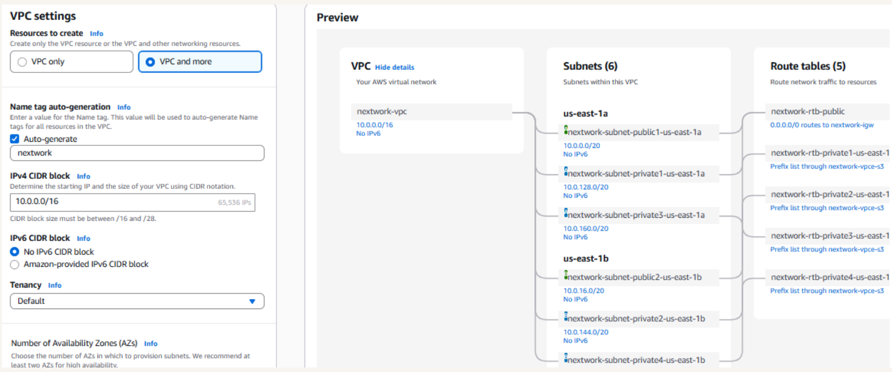
nextwork.org

Launching VPC Resources

CH Arpita Chowdhury



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a Virtual Private Cloud that lets you create an isolated, customizable network environment within AWS. It is useful because it gives you full control over your networking including IP address ranges, subnets, route tables, gateways, and security groups — so you can securely run resources (like EC2 instances, databases, and applications) in the cloud.

How I used Amazon VPC in this project

I used Amazon VPC to design a secure network for my project by creating both public and private subnets, attaching an internet gateway for public access, and configuring route tables and security groups to control traffic. This allowed me to place resources like web servers in the public subnet and databases in the private subnet, ensuring both accessibility and security.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was using VPC and more were all other resources like route table, security group, internet gateway and network ACL are all created simultaneously and not needed to be created seperately.

This project took me...

This project took me 2 hours.

Setting Up Direct VM Access

Directly accessing a virtual machine means connecting to your EC2 instance itself (for example, via SSH for Linux or RDP for Windows) using its IP address and key pair, without going through a load balancer, bastion host, or other intermediate service.

SSH is a key method for directly accessing a VM

SSH traffic means the secure, encrypted communication exchanged between a client and a server using the SSH (Secure Shell) protocol, typically over TCP port 22.

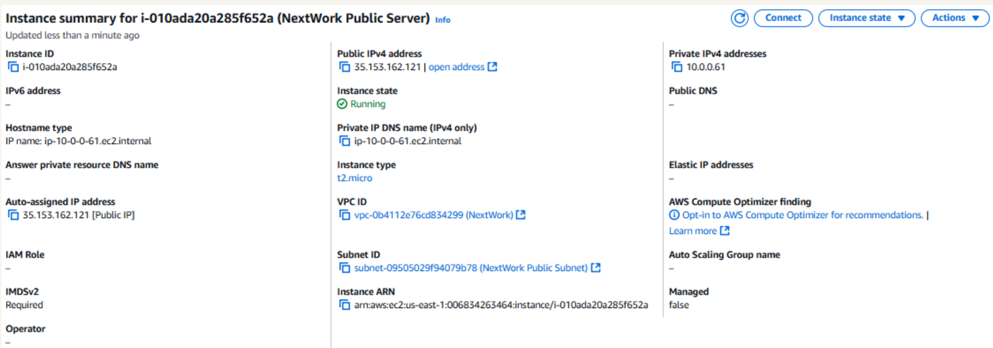
To enable direct access, I set up key pairs

A key pair are used for secure login to EC2 instances without needing a password.

A private key's file format means the type of file extension used to store the key material (such as .pem or .ppk). My private key's file format was .pem.

Launching a public server

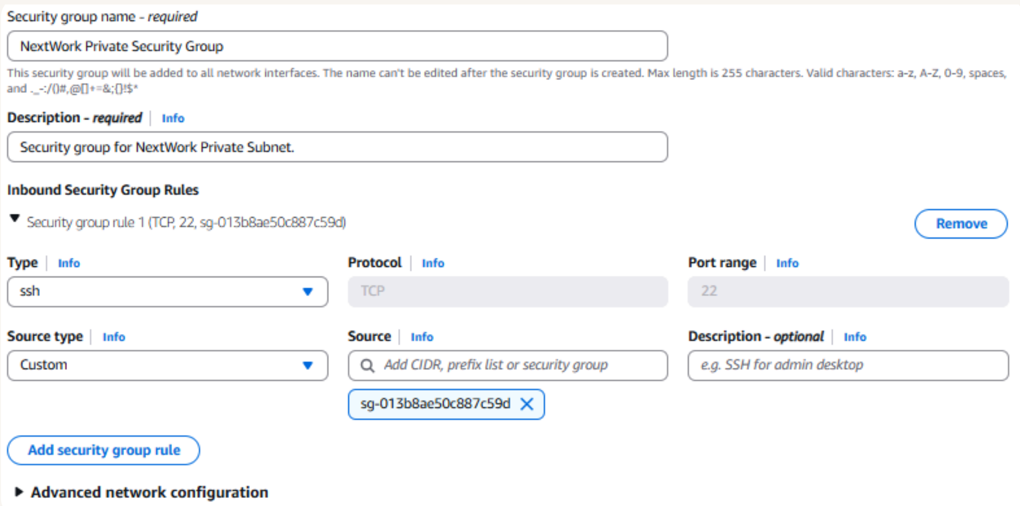
I had to change my EC2 instance's networking settings by selecting the NextWork VPC for VPC, NextWork public subnet for subnet, existing security group for security group and NextWork secuiry group for common security group.



Launching a private server

My private server has its own dedicated security group because it requires stricter access controls, allowing only specific internal traffic (for example, from the public server or a load balancer) while blocking direct access from the internet.

My private server's security group's source is another security group (the public server’s SG), which means only instances that belong to that source security group are allowed to initiate SSH connections into my private server.

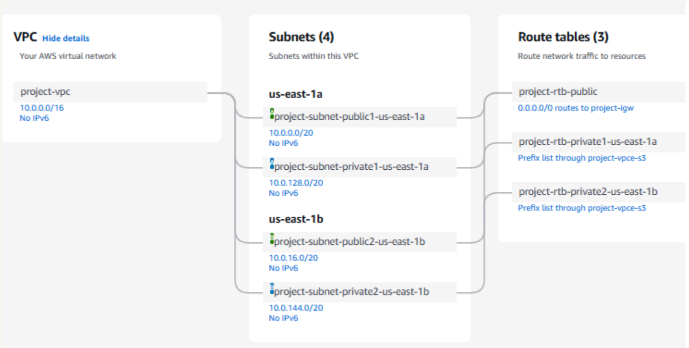


Speeding up VPC creation

I used an alternative way to set up an Amazon VPC! This time, I used VPC and more in the VPC console wizard, which automatically created a VPC with public and private subnets, route tables, and optional NAT gateways across two Availability Zones. This setup follows AWS best practices for high availability and redundancy, making the network production-ready from the start.

A VPC resource map is a visual diagram that shows all the networking resources inside a Virtual Private Cloud (VPC) such as subnets, route tables, security groups, gateways, and instances and how they are connected to each other.

My new VPC has a CIDR block of 10.0.0.0/16. It is possible for my new VPC to have the same IPv4 CIDR block as my existing VPC because VPCs are isolated virtual networks, and their IP ranges do not overlap unless I explicitly connect them (for example, through VPC peering or a Transit Gateway).



VPC Hide details
Your AWS virtual network
project-vpc
10.0.0.0/16
No IPv6
Subnets (4)
Subnets within this VPC
us-east-1a
project-subnet-public1-us-east-1a
10.0.0.0/20
No IPv6
project-subnet-private1-us-east-1a
10.0.128.0/20
No IPv6
us-east-1b
project-subnet-public2-us-east-1b
10.0.16.0/20
No IPv6
project-subnet-private2-us-east-1b
10.0.144.0/20
No IPv6
Route tables (3)
Route network traffic to resources
project-rtb-public
0.0.0.0/0 routes to project-igw
project-rtb-private1-us-east-1a
Prefix list through project-vpce-s3
project-rtb-private2-us-east-1b
Prefix list through project-vpce-s3

Speeding up VPC creation

Tips for using the VPC resource map

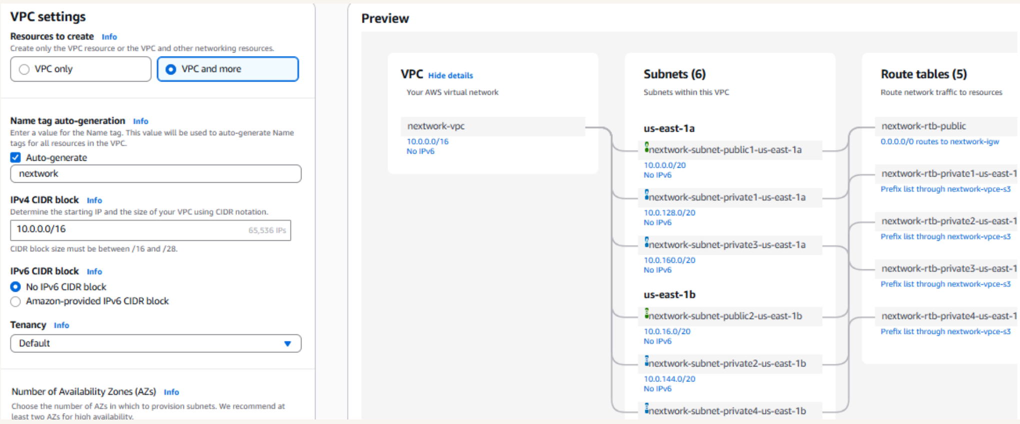
When determining the number of public subnets in my VPC, I only had two options: 0 and 2. This was because AWS' best practice advice is built into the VPC wizard. If I select two Availability Zones, the wizard automatically creates a public subnet in each one. This ensures redundancy (having backups in different places) and high availability (keeping resources accessible even if one AZ goes down). Just one public subnet would not provide this reliability, so the option isn't offered. The wizard also limits me to two public subnets for simplicity, but I can always add more manually later if needed (up to 200 in a VPC).

The setup page also offered to create NAT gateways, which are managed AWS services that let private subnets access the internet securely without exposing those resources to inbound internet traffic.

CH

Arpita Chowdhury
NextWork Student

nextwork.org



nextwork.org

The place to learn & showcase your skills

Check out nextwork.org for more projects